KAIZEN MATHS

Lesson starter

Inequalities

Advanced Linear Inequalities

Name _____

S1

$-2x + 2 \leq 8x - 2$

S2

$2(x - 4) < 27$

S3

$-8x - 2 > -17$

S4

$5x - 9 > -6x - 10$

Show your working

KAIZEN MATHS

Lesson starter

Inequalities

Advanced Linear Inequalities

Name _____

S1

$-2x + 2 \leq 8x - 2$

S2

$2(x - 4) < 27$

S3

$-8x - 2 > -17$

S4

$5x - 9 > -6x - 10$

ALG-1962-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Inequalities

Advanced Linear Inequalities

Name _____

E1

$-9(x-3) \leq 17$

E2

$7-2x \geq 20$

E3

$-4x+11 \leq 3x+4$

KAIZEN MATHS

Exit ticket

Inequalities

Advanced Linear Inequalities

Name _____

E1

$-9(x-3) \leq 17$

E2

$7-2x \geq 20$

E3

$-4x+11 \leq 3x+4$

ALG-1962-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Inequalities

Advanced Linear Inequalities

Name _____

E1

$-9(x-3) \leq 17$

E2

$7-2x \geq 20$

E3

$-4x+11 \leq 3x+4$

ALG-1962-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Inequalities

Advanced Linear Inequalities

Name _____

E1

$-9(x-3) \leq 17$

E2

$7-2x \geq 20$

E3

$-4x+11 \leq 3x+4$

ALG-1962-L2 · E

Show your working

Teacher answers & working

Inequalities · Advanced Linear Inequalities

ALG-1962-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

$-2x + 2 \leq 8x - 2$

Answer:

$x \geq \frac{2}{5}$

Working:

Step 1: Start with the original inequality.

$-2x + 2 \leq 8x - 2$

Step 2: Use inverse operations to move terms away from x.

$-10x + 2 \leq -2$

$-10x \leq -4$

Step 3: Divide by a negative, so reverse the inequality sign.

$x \geq \frac{-4}{-10}$ (flip)

Step 4: Write the simplified solution for x.

$x \geq \frac{2}{5}$

S2

$2(x - 4) < 27$

Answer:

$x < \frac{35}{2}$

Working:

Step 1: Start with the original inequality.

$2(x - 4) < 27$

Step 2: Use inverse operations to move terms away from x.

$x - 4 < \frac{27}{2}$

$x - 4 < \frac{27}{2}$

Step 3: Write the simplified solution for x.

$x < \frac{35}{2}$

Teacher answers & working

Inequalities · Advanced Linear Inequalities

ALG-1962-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

$-8x - 2 > -17$

Answer:

$x < \frac{15}{8}$

Working:

Step 1: Start with the original inequality.

$-8x - 2 > -17$

Step 2: Use inverse operations to move terms away from x.

$-8x > -17 + 2$

Step 3: Divide by a negative, so reverse the inequality sign.

$x < \frac{-17 + 2}{-8}$ (flip)

Step 4: Write the simplified solution for x.

$x < \frac{15}{8}$

S4

$5x - 9 > -6x - 10$

Answer:

$x > \frac{-1}{11}$

Working:

Step 1: Start with the original inequality.

$5x - 9 > -6x - 10$

Step 2: Use inverse operations to move terms away from x.

$11x - 9 > -10$

$11x > -1$

$x > \frac{-1}{11}$

Step 3: Write the simplified solution for x.

$x > \frac{-1}{11}$

Teacher answers & working

Inequalities · Advanced Linear Inequalities

ALG-1962-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

$-9(x-3) \leq 17$

Answer:

$x \geq \frac{10}{9}$

Working:

Step 1: Start with the original inequality.

$-9(x-3) \leq 17$

Step 2: Divide by a negative, so reverse the inequality sign.

$x-3 \geq \frac{17}{-9}$ (flip)

$x-3 \geq \frac{-17}{9}$

Step 3: Write the simplified solution for x.

$x \geq \frac{10}{9}$

E2

$7-2x \geq 20$

Answer:

$x \leq \frac{-13}{2}$

Working:

Step 1: Start with the original inequality.

$7-2x \geq 20$

Step 2: Use inverse operations to move terms away from x.

$-2x \geq 13$

Step 3: Divide by a negative, so reverse the inequality sign.

$x \leq \frac{13}{-2}$ (flip)

Step 4: Write the simplified solution for x.

$x \leq \frac{-13}{2}$

Teacher answers & working

Inequalities · Advanced Linear Inequalities

ALG-1962-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

$-4x + 11 \leq 3x + 4$

Answer:

$x \geq 1$

Working:

Step 1: Start with the original inequality.

$-4x + 11 \leq 3x + 4$

Step 2: Use inverse operations to move terms away from x.

$-7x + 11 \leq 4$

$-7x \leq -7$

Step 3: Divide by a negative, so reverse the inequality sign.

$x \geq \frac{-7}{-7}\ (\text{flip})$

Step 4: Write the simplified solution for x.

$x \geq 1$